

Car Data Management Module

The Car Data Management Module is responsible for storing, organizing, and managing all car-related information in the Smart Car Recommendation System. It acts as the backbone of the application by maintaining accurate and structured data required for recommendations, comparisons, and filtering. This module handles detailed car data such as brand, model, price, fuel type, mileage, engine specifications, safety ratings, features, and images. All this information is stored in a database in a well-structured format for efficient retrieval. An Admin functionality is included in this module, allowing authorized users to add, update, or delete car records. This ensures that the system always contains up-to-date and relevant data. The module also supports image management, where multiple images (front, rear, side, interior) are stored and displayed for each car. This enhances user experience by providing a better visual understanding of the vehicle. Data validation is an important part of this module. It ensures that all inputs such as price, mileage, and specifications are accurate and consistent before storing them in the database. This module is tightly integrated with other components such as the recommendation engine, comparison module, and filtering system, ensuring seamless data flow across the platform. Technologies Used: - HTML: For creating admin input forms and data display pages. - CSS: For styling and making the interface responsive. - JavaScript: For dynamic form handling and validation. - Python (Flask): For backend logic, API creation, and data processing. - Database (MongoDB/MySQL): For storing and managing all car-related data. Overall, the Car Data Management Module ensures efficient handling of large volumes of car data, enabling accurate recommendations and smooth system performance.